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import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Read the uploaded image
img = cv2.imread('/content/image.jpg')

# Convert image to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Laplacian mask with negative center coefficient
laplacian_mask = np.array([
    [0, 1, 0],
    [1, -4, 1],
    [0, 1, 0]
], dtype=np.float32)

# Apply Laplacian mask
laplacian = cv2.filter2D(gray, cv2.CV_32F, laplacian_mask)

# Sharpen the image
sharpened = gray.astype(np.float32) - laplacian

# Convert to 8-bit
sharpened = cv2.convertScaleAbs(sharpened)

# Display Original Image
print("Original Image")
cv2_imshow(img)

# Display Grayscale Image
print("Grayscale Image")
cv2_imshow(gray)

# Display Laplacian Image
print("Laplacian Image")
cv2_imshow(cv2.convertScaleAbs(laplacian))
```

```
# Display Sharpened Image  
print("Sharpened Image")  
cv2_imshow(sharpened)
```





Grayscale Image

